

	Case Name: Beer Tasting	Sector	IT
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule Risk Analysis	Schedule with resources Schedule with costs Random simulation One of nine std. scenarios User defined distributions
Submitted by	Matilde Casado and Margarida Antunes		
Date	2015	Project Control	Automatic tracking
File Name	C2015-18 Beer Tasting		Tracking based on user input

1. Project description

Project authenticity

The Tax Return System (2) project involves analysis, design, development, testing, implementation, and post-go-live support to enhance tax processing and system efficiency.

The project consists of activity, resource and cost data that were obtained directly from the actual project owner.

2. Project properties

2.1. Baseline Schedule

General	
# Activities	21
Planned Duration (PD)	128 days*
Budget At Completion (BAC)	39.450 €
Renewable Resources	6
Consumable Resources	-

* standard eight-hour working days

Network topology	
Serial/Parallel (SP)	10%
Activity Distribution (AD)	66%
Length of Arcs (LA)	16%
Topological Float (TF)	11%

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	15.5	15.2	-0.6
CRI-rho	29.6	17.5	0.1
CRI-tau	38.5	40.5	-0.9

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	31.4	7.2	-2.1
CRI-rho	31.4	7.2	-2.1
CRI-tau	31.4	7.2	-2.1

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	14.5	28.3	-2.2
SI	25.4	28.0	-1.0
SSI	7.2	19.5	-3.1
CRI-r	9.7	15.7	-3.6
CRI-rho	23.9	22.1	-0.8
CRI-tau	37.2	42.2	-0.8

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy			Simulated EAC accuracy		
method - PF	MAPE [%]	MPE [%]	method (PF)	MAPE [%]	MPE [%]
PV - 1	212.7	-212,7	1	1.5	1.3
PV - SPI	134.0	-134.0	CPI	1.7	1.7
PV - SCI	127.6	-127.6	SPI	8.6	8.6
ED - 1	697.4	-697.4	SPI(t)	9.9	9.9
ED - SPI	134.0	-134.0	SCI	8.7	8.7
ED - SCI	129.2	-129.2	SCI(t)	9.9	9.9
ES - 1	171.1	-171.1	0.8 CPI + 0.2 SPI	5.5	5.5
ES - SPI(t)	25.1	4.1	0.8 CPI + 0.2 SPI(t)	8.2	8.2
ES - SCI(t)	25.4	4.5			

According to the MAPE values¹ the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 3 tracking periods with a length of approximately 1 month. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

¹ The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	6211.9	-6308.1	-128.0	1.3	0.7	0.6	0.8
std dev	6952.3	5504.1	122.0	0.2	0.3	0.4	0.2
final	14070.0	0.0	0.0	1.6	1.0	1.0	1.0

2.3.3.2. Time forecasting

PD	128 days	Real Duration	128 days	Late	0%
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EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	161.7	0.6	8.4	-8.4
PV - SPI	162.0	0.0	8.5	-8.5
PV - SCI	162.3	0.6	8.6	-8.6
ED - 1	163.0	1.0	8.8	-8.8
ED - SPI	164.0	1.0	9.0	-9.0
ED - SCI	165.0	1.0	9.3	-9.3
ES - 1	166.0	1.0	9.6	-9.6
ES - SPI(t)	166.7	0.6	9.7	-9.7
ES - SCI(t)	167.0	0.0	9.8	-9.8

2.3.3.3. Cost forecasting

BAC	39.450 €	Real Cost	25.380 €	Under Budget	-35.7%
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EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	33238.2	6952.3	10.3	-10.3
CPI	30018.0	4468.8	6.1	-6.1
SPI	54323.8	32263.7	38.0	-38.0
SPI(t)	79598.9	61441.5	71.2	-71.2
SCI	47927.2	27255.1	29.6	-29.6
SCI(t)	69284.4	52864.6	57.7	-57.7
0.8 CPI + 0.2 SPI	32127.9	6623.8	8.9	-8.9
0.8 CPI + 0.2 SPI(t)	32752.3	7174.6	9.7	-9.7